

WireDrum: An Interpersonal and Multimodal Skill-Sharing System for an Augmented Drumming Experience

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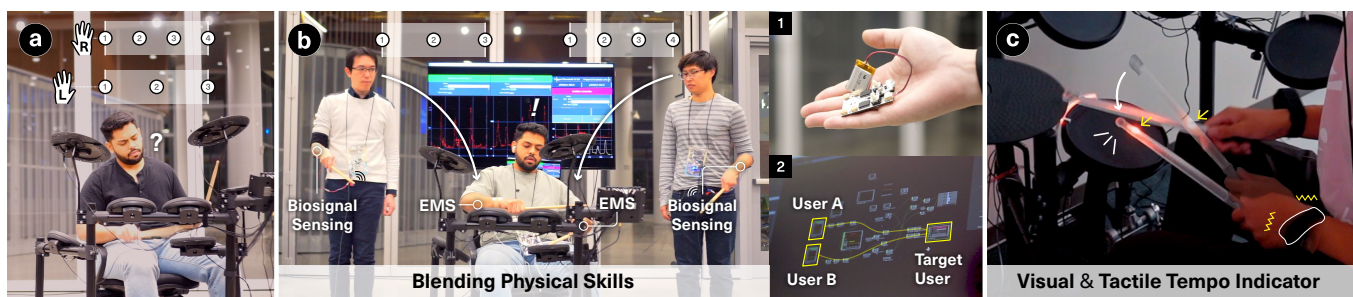


Figure 1: (a) Learning a complex physical skill (e.g., drumming) requires time and effort. (b) Our WireDrum system, which comprises wearable biosignal sensors and electrical muscle stimulators, integrates the actions of two instructors into a target user's muscles, enabling them to perform complex skills with reduced teaching effort. (c) After this, the target user can further refine their timing for hitting the drum on their own through visual and tactile feedback provided by a custom drumstick.

Abstract

We present WireDrum — a synthetic, multimodal experience that enables motor skill transfer and learning among users. The system consists of multiple custom biosignal-sensing modules, a custom electrical muscle stimulator, a visual and tactile tempo indicator, and muscle-topology configuration software. By integrating the muscle activities of multiple instructors into a target user's muscles, the user can perform complex physical skills beyond their original ability. Simultaneously, instructors can divide and reduce their cognitive load related to teaching and performing the task. We demonstrate a shared somatic and interpersonal drumming experience, where a visitor can play a cognitively demanding rhythm that is challenging for novices to perform alone by sharing the visitor's body with other visitors.

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1 Introduction

Developing dexterous physical skills, such as drumming, is challenging and requires mastering multidimensional aspects, including physically coordinating both hands. It is well known that providing **haptic feedback to a music learner** enables higher learning efficacy. This has been achieved by different types of force-feedback, such as exoskeletons with motors [Grindlay 2008], pneumatic actuators, vibrotactile motors, and even ones using electrical muscle stimulation (EMS). EMS is one of the effective options for providing fast force-feedback since it can activate muscles and cause physical body movements while only requiring tiny batteries and electrodes; thus, the HCI community has employed this technique in learning applications [Miura and Hamanaka 2021; Nishida et al. 2019]. Interestingly, there is also substantial neuroscientific evidence that supports the role of **interpersonal haptic communication in motor learning**. Ganesh et al. found that when two users executed a shared motor task together, they achieved higher performance when compared with doing the task alone [Ganesh et al. 2014].

2 WireDrum: Synthetic Somatosensation

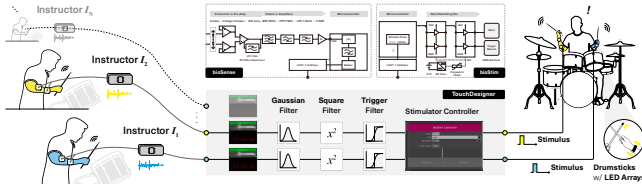


Figure 2: Schematics of WireDrum System

By synthesizing findings on the efficacy of providing haptic feedback to musicians via EMS and the role of interpersonal haptic communication in physical skill learning, we propose *WireDrum* — an interactive system that enables synthetic somatosensory experiences to enhance skill communication and learning. It consists of 1) custom biosignal-sensing modules on instructors, 2) a custom EMS device on a target user, and 3) muscle-topology configuration software. This setup integrates multiple users' muscle activities and unique physical skills into a single enhanced skill, while reducing the physical and cognitive teaching load for each instructor. To further refine their learning with a higher sense of agency and enhanced tactile feedback [Fuehrer et al. 2022], we also designed drumsticks that provide visual and tactile feedback. It uses multiple LEDs in a flowing sequence, allowing the user to predict the performance timing while providing vibrotactile cues on the wrist.

Visitor Experience. As an example, we demonstrate a **somatic and interpersonal drumming experience**: 1) A visitor will be asked to play a cognitively demanding rhythm — specifically, a 4-beat rhythm with the right hand and a 3-beat rhythm with the left hand, which cannot be performed by a single user (Figure 1a). 2) The visitor's hands will be connected to those of other visitors (acting as instructors) through sensing and stimulation electrodes. This setup will allow each instructor-visitor to control one of the target user's hands, enabling them to collaboratively perform the complex rhythm with a significantly reduced teaching load (Figure 1b). 3) After this, the visitor can also try the tempo indicator, which enables self-learning of correct timings with less sense of intervention, fostering further skill acquisition (Figure 1c).

3 System Implementation

We designed *bioSense*, a **wearable 2-ch EMG device** (60x30x15 mm) comprising an INA2128 differential amplifier and multiple low/high-pass filters, to capture muscle activities from the instructor during drumming. The EMG signal is processed with an absolute-value filter and a median filter, and then transmitted to our muscle-topology software in *TouchDesigner*, which allows end-users to easily change muscle topology visually (Figure 1b-2). The signal is further filtered by using a Gaussian filter and is used to generate a preemptive trigger signal for our **wearable 4-ch EMS device** (Figure 1b-1), *bioStim* (based on [Nishida et al. 2019]). This device delivers EMS pulses to the visitor, enabling them to replicate the instructors' drumming motion in real-time. It consists of a 3.7 V LiPo battery, a DC-DC converter (35 V), gate-switching devices (TI, L293DD), an Arduino Nano ESP32 microcontroller, and a pair of electrodes.

The microcontroller generates pulse signals (pulse width: 0–500 μ s, frequency: 30–100 Hz) that control the gate-switching devices, delivering EMS pulses to the electrodes. A **wrist vibro band** provides a 200 Hz sine-wave vibration using a vibromotor (ACTUATOR 639893, Foster Electric Company). The signal is sent from a computer through an audio interface (Sound Blaster PLAY! 3, Creative) and an amplifier (AP05mk2, Fostex). An **LED strip** (Figure 1c), containing eight LEDs attached to a drumstick, is controlled by a microcontroller (ESP32-DevKitC), lighting in sequence from the user's grip end toward the tip. When the LED at the tip turns on, it indicates the correct timing to hit the drum kit (Yamaha DTX432K).

4 Pilot Study: Teaching Load Evaluation

We conducted a preliminary study to observe potential changes in teaching load. In the **single** condition, one participant performs a 3-beat rhythm with the right hand and a 4-beat rhythm with the left hand simultaneously to teach a learner experimenter. In the **dual** condition, the participant and a teacher experimenter each play one rhythm — 4-beat or 3-beat — using one hand, while coordinating with each other. NASA-TLX was employed to assess teaching load. Participants ($n=8$) completed 2 minutes of rhythm training, then tried each condition (counter-balanced), and finally completed the questionnaire. As a result, the dual teaching setup mitigated mental (-2.67), physical (-1.67), and temporal (-1.33) demands, along with effort (-2.33) and frustration (-1.13), while enhancing success (+1.63) compared to the single condition.

5 Conclusion

We present *WireDrum* — a synthetic, multimodal experience that enables motor skill transfer and learning among users. The system consists of multiple custom biosignal-sensing modules, a custom EMS device, a visual and tactile tempo indicator, and muscle-topology configuration software. We demonstrate a shared somatic and interpersonal drumming experience using the system.

Acknowledgments

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